

Week 10 – Workshop Plan

Finite Differences

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Workshop Overview

This workshop builds intuition for finite difference methods through physical reasoning and interactive activities. Students first derive the heat equation from energy conservation, then experience how numerical models work by acting as nodes that update temperature based on their neighbours. The workshop then explores how boundary conditions represent geological assumptions and how numerical stability reflects limits on how heat can physically propagate. By the end, students should understand that finite difference methods are simply local energy balance, that heat flow emerges from neighbour interactions, and that both boundary conditions and stability have direct physical meaning.

Learning Objectives

Students will:

- Understand finite difference equations as local statements of energy conservation.
- Develop intuition for how numerical solutions evolve through neighbour interactions.
- Interpret boundary conditions as geological assumptions.
- Understand numerical stability as a limit on physical information transfer.

0:00–0:30 From Energy Conservation to Finite Differences

Concept

Finite difference equations arise from applying conservation of energy to small rock volumes.

- Nodes represent local rock volumes.
- Heat flux enters and leaves each node.
- Heat production adds energy locally.

Activity: From Conservation of Energy to Finite Differences

The equation is *not a numerical trick*. The Laplacian as a measure of curvature (difference from neighbours). This is just bookkeeping of energy.

0:30–1:05 Finite Differences as a Physical Process

Concept

Numerical models evolve through local interactions between neighbouring nodes.

Activity: Human Finite Difference Solver

Discuss:

- smoothing of temperature
- spreading of heat
- role of gradients

1:05–1:25 Boundary Conditions as Geological Assumptions

Concept: Boundary conditions determine what physical system the model represents.

- Fixed temperature (Dirichlet)
- Fixed heat flux (Neumann)

Activity: Boundary Conditions and Geological Meaning

1:25–1:35 Break

1:35–2:00 When Analytic Solutions Break Down

Concept

- Analytic solutions require simplifications:
 - constant properties
 - simple geometries
 - simple boundary conditions
- Real geology violates these assumptions.

Discussion prompts

- What makes geological systems difficult to solve analytically?
- Why do irregular boundaries force numerical approaches?

Your Iberia exercise on the problem has several issues that make it impossible to develop an analytic form, including variable sediment thickness, lateral structure, varying boundary condition.

2:00–2:40 Numerical Stability and Information Transfer

Concept Stability limits impose a “speed limit” on thermal information:

$$\Delta t < \frac{\Delta x^2}{2\kappa}$$

- Heat propagates gradually through diffusion.
- Numerical schemes must respect this.

Activity: Stability, Time Stepping, and Information Transfer

- small vs large time steps
- what happens when updates are too large
- stability = causality
- information cannot skip nodes

Instability is not just numerical—it is physically impossible behaviour.

2:40–3:00 Wrap-Up and Synthesis

Key connections

- Finite differences = local energy balance + neighbour interaction
- Boundary conditions = geological assumptions
- Stability = physical constraint on information transfer

Discussion

- How does the numerical method compare to analytic solutions?
- What advantages do numerical models provide?
- Where might these methods be applied beyond heat flow?

Preview

- Extension to 2-D systems
- Application to real datasets (Iberia Abyssal Plain)